

SQA · BENGKEL AMALI 2 HARI

KRISA 2.0 (JDN) ALIGNED · PHASE 05 OF 05, SUSTAIN · FINAL MODULE BEFORE CAPSTONE

MODULE 08

Change Management, Monitoring & Continuous Quality

Control what changes. Watch what runs. Never stop improving.



15:15 - 16:30

SESSION TIME



1h 15m

DURATION



SUSTAIN

PIPELINE PHASE

Sample module reference · ShopFast checkout continuity project

Jadual slot: Modul 8: Perubahan, Pemantauan dan Kualiti Berterusan

Run sheet / session timing

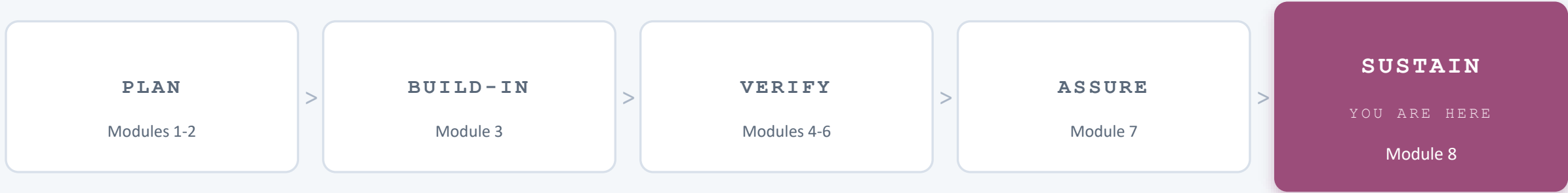
Hari 2, 15:15 hingga 16:30 · 75 minit · kira-kira 80% amali



Masa	Segmen	Tempoh
15:15 hingga 15:35	Konsep	20 minit
15:35 hingga 16:10	Makmal	35 minit
16:10 hingga 16:25	Bengkel AI W08	15 minit
16:25 hingga 16:30	Rumusan	5 minit

Where Module 8 fits, the last phase before capstone

The audited, documented release from Module 7 now goes live. Module 8 keeps it healthy, and feeds what's learned back into the next cycle.



Coming in from Module 7: a signed-off Audit Report, CAPA log and Test Summary Report with a GO / conditional release call.

Module 8 asks: now that it's live, how do we change it safely, know the moment it breaks, and keep getting better?

Closing the loop: PDCA/Kaizen feeds monitoring insights back into Module 1's quality mindset, then the Capstone brings all nine activities together.



LEARNING OBJECTIVE

Sustain quality after deployment through controlled change and production monitoring, so the system stays healthy and keeps improving.

15:15-16:30 · Change Management + Monitoring

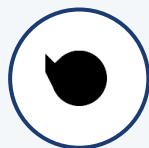
What you leave the room with

Five artifacts that together prove a change was safe, and the system is being watched.



Change Request + Impact Analysis

Logged change, mapped to affected test cases via the RTM.



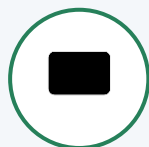
Rollback Plan

A minimal regression suite and a documented way back.



CI Quality Gate

Automation tests wired into [GitHub Actions](#) before merge.



Metrics Dashboard

Defect Density, pass rate, MTTR and uptime, with alert thresholds set.

Key topics covered



Change Request · Impact Analysis

What's changing, and what does it touch?



Change Advisory Board (CAB)

Who decides go / no-go on a change



Rollback · Regression Planning

The safety net if something breaks



Logging / Monitoring / Alerting

Seeing problems before users report them



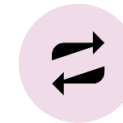
Observability

Logs, metrics and traces working together



SLA · SLI · SLO

Promises, measurements, and targets

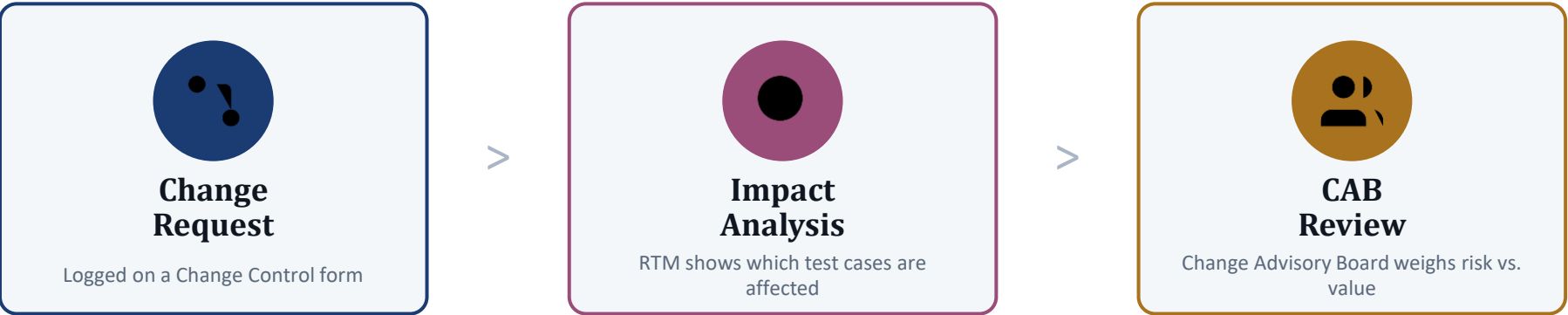


PDCA · Kaizen

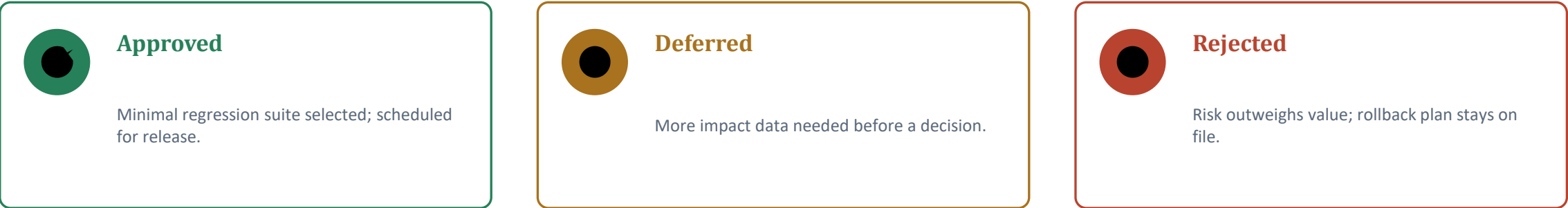
Closing the loop, plan, do, check, act

From request to decision, the change control path

Every change, even 'just add discount codes to checkout', walks this path before it touches production.



CAB DECISION FORKS INTO THREE OUTCOMES



Kawalan pindaan: dari permohonan ke keputusan

Change control: from request to decision (PPrISA)

PPrISA 2.0 4.2.3 c) dan 5.4.2 i a); KRISAv2 Bab 1 1.6.5; Bab 8 8.3.3: impak direkod dalam Log Perubahan dan melalui sprint perubahan

1

Mohon pindaan

PPrISA10 Borang Pindaan

2

Analisis impak

RTM: skop, masa, kos, kualiti

3

Keputusan CCB

Lembaga Kawalan Pindaan

4

Rekod

PPrISA11 Log Penyelesaian Pindaan

5

Laksana dan uji regresi

Quality Gate CI/CD

6

Lapor status

PPrISA16 / PPrISA17

Diluluskan

Kemas kini RTM dan pelan ujian, jalankan regresi minimum, sediakan pelan rollback, kemas kini Laporan Release.

Ditolak atau ditangguh

Rekod sebab dalam PPrISA11, maklumkan pemohon, masukkan semula ke backlog jika perlu.

Contoh ShopFast

CR-001: tambah kod diskaun baharu RAYA26 pada checkout. Impak: REQ-CHK-02 dan REQ-CHK-03, 7 kes ujian regresi dipilih dari RTM.

Seeing inside a running system

Logging, monitoring and alerting are three different jobs. Observability is what you get when all three talk to each other.



Logs

What happened, in detail, the ShopFast checkout request that threw a 500.



Metrics

How the system is trending, pass rate, latency, error rate over time.



Traces

Where the time went, one request's path across every service it touched.



Alerting sits on top of all three: a threshold breach on any metric, log pattern, or trace latency fires a notification, before a customer has to file a ticket.

SLA, SLI, SLO, three letters, three different jobs

They nest inside each other: a measurement feeds a target, and a target backs a promise.

SLA, Service Level Agreement

The promise made to the customer (e.g. 99.9% uptime/month, with penalties if missed).

SLO, Service Level Objective

The internal target the team aims for, stricter than the SLA (e.g. 99.95% uptime).

SLI, Service Level Indicator

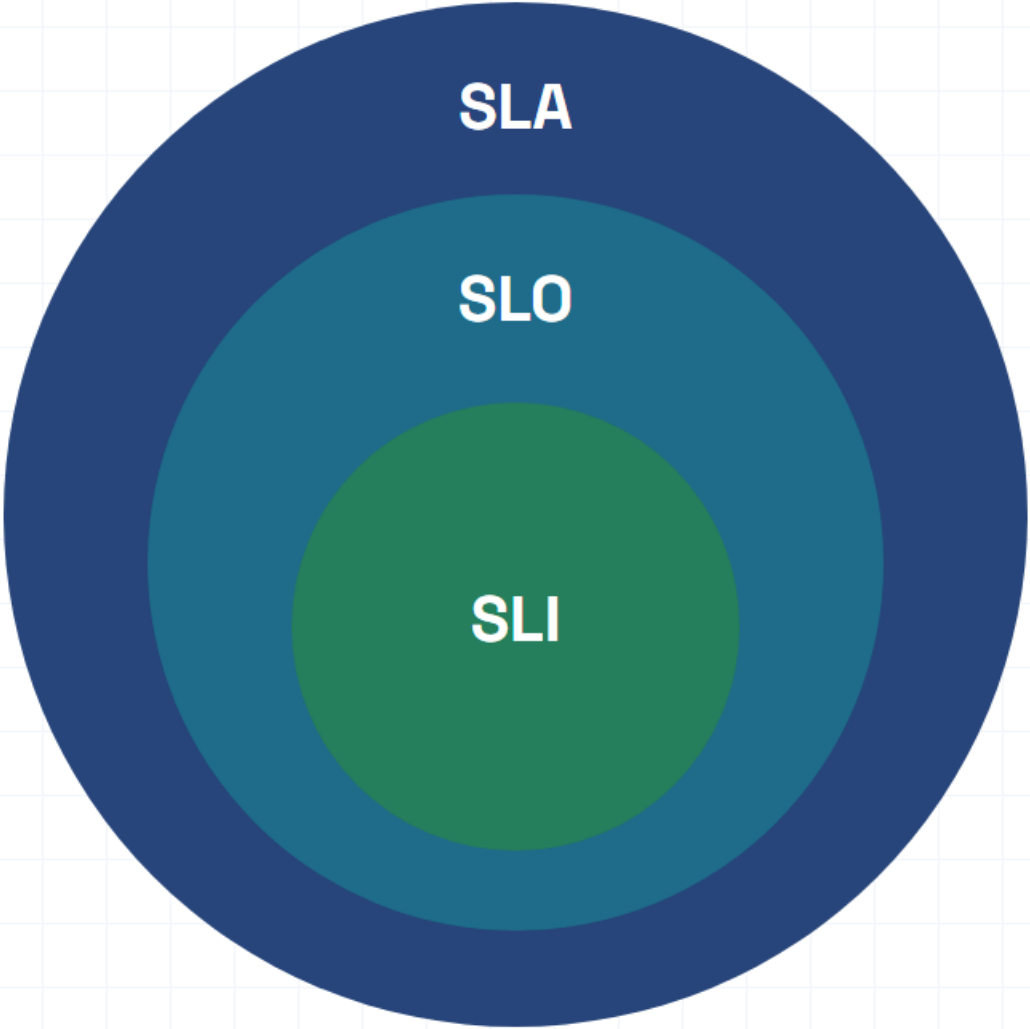
The actual measurement, read straight off the dashboard (e.g. 99.97% this week).

Reading it live: SLI (99.97%) meets SLO (99.95%) with room to spare, so the SLA promise (99.9%) to the customer stays safely kept.

SLA, SLO dan SLI: janji, sasaran dan ukuran

SLA, SLO and SLI: promise, target and measurement

Pemantauan selepas pelaksanaan: KRISAv2 Bab 8 (APM Tools dan Log Pemantauan dalam serahan Pelaksanaan)



	Maksud	Contoh ShopFast
SLA	Perjanjian dengan pelanggan, ada penalti	Ketersediaan 99.5% sebulan dalam kontrak
SLO	Sasaran dalaman, lebih ketat dari SLA	99.9% checkout berjaya, p95 respons bawah 800 ms
SLI	Ukuran sebenar dari sistem	Checkout berjaya / jumlah cubaan, dari log dan metrik

Change, Monitoring and Continuous Quality

Keeping quality under control after the first release, and improving the process that produces it

01 Software configuration management activities

02 SCM vocabulary: CI, baseline, CR, CCB, ECP

03 Controlling one change end to end

04 Software evolution and Lehman's laws

05 Review checklist for maintenance changes

06 Process and product quality loop

07 Quality improvement and Six Sigma

08 Prevention culture and death march warning signs

Software configuration management activities

SCM makes sure every product is identified and every change to it is tracked

**Baseline: approved
version under control**

Identification

Name and version
every configuration
item and define the
baseline

Change identification

Record each
change made
against the
baseline

Change control

Evaluate, approve
or reject changes
through the CCB,
keep revision
history

Status accounting

Report what
changed, where,
and the state of
every item

Audit and preservation

Verify the build
matches its records
and keep archives
safe

ShopFast: v1.0.0 is the baseline; CR-001 (RAYA26) is a change request; the CCB decision, the changed files and the new tag v1.1.0 are the SCM record.

SCM activities in practice

Five activities from the source, plus the configuration audit named in SCM standards

Activity	What it does	ShopFast example
Baseline identification	Identify the initial approved state of each product item	Tag release 1.0: SRS v1.0, checkout code commit, test procedures v1.0
Change identification	Identify every change made against the baseline	Every commit and document revision cites a change request ID
Change control	Evaluate, approve and document changes: revision history, change summaries, tools	CCB approves raising the maximum quantity from 10 before any code is written
Status accounting	Report changes to others; monitor completeness of project archives	Weekly report of open, approved, implemented and verified change requests
Preservation	Keep the software products safe and retrievable	Protected repository and backups; release 1.0 can be rebuilt exactly
Configuration audit	Check the product matches its documents and recorded changes	Before release, compare build contents with the version description

SCM vocabulary every SQA engineer uses

Configuration item (CI)

Any work product placed under configuration control: code, SRS, design, test procedures, build scripts.

Baseline

A formally reviewed and approved version of one or more CIs, changed only through formal change control.

Change request (CR)

A documented request to change a baselined item, with its reason, impact and priority.

CCB

Configuration or change control board: approves, defers or rejects changes. The SCCB in the bug workflow.

ECP

Engineering change proposal: a formal proposal to change a contracted baseline, usually with cost and schedule impact.

Traceability

Links each change to its request, code commits, tests and release, so any change can be explained later.

Controlling one change from request to release

Example: the business asks to lower ShopFast free shipping from RM200 to RM150



No code is touched before the CCB decision. A change that skips steps is an audit finding waiting to happen.

Software evolution: change is the normal state

Leveson's notes: development costs are only the tip of the iceberg

Continuing change

Belady and Lehman: software that is used will continually change.

Growing disorder

Belady and Lehman: software becomes increasingly unstructured as it is changed, unless effort is spent to restructure it.

Where maintenance goes

As cited by Leveson: about 20 percent error correction, 20 percent adaptation, 60 percent enhancements.

Where fielded errors start

Most errors found in fielded software stem from requirements, not from code.

Leveson's law

Introducing computers will not reduce personnel numbers or costs: a caution against over-promising automation.

ShopFast implication

Plan for promo rules, payment gateways and shipping fees to change. Budget maintenance and regression testing from day one.

Review checklist: maintenance changes

Apply to every change request before it is merged into the baseline

- ☐ Have side effects of the change been considered, including code that calls the changed parts?
- ☐ Have appropriate formal technical reviews been held on the changed items?
- ☐ Has the change request been documented, evaluated and approved before work started?
- ☐ Has regression testing covered the affected areas, not just the changed lines?
- ☐ Has the impact on requirements, design, tests and user documents been identified?
- ☐ ShopFast: after changing the shipping threshold, do promo code and order total tests still pass?
- ☐ Once made, has the change been documented and reported to interested parties?
- ☐ Is the new baseline recorded and the version description updated?

Most maintenance defects are side effects: the fix works, but something nearby breaks.

Process quality and product quality

A good process usually gives a good product, but software is designed, not manufactured

Factory approach

Define process

Develop product

Assess quality

Quality OK?

If not OK, improve the process and repeat; if OK, standardise it.

Works for manufacturing, where the same product is made many times. Weak for software, where every product is new.

Creative approach

Find good products

Study their processes

Generalise

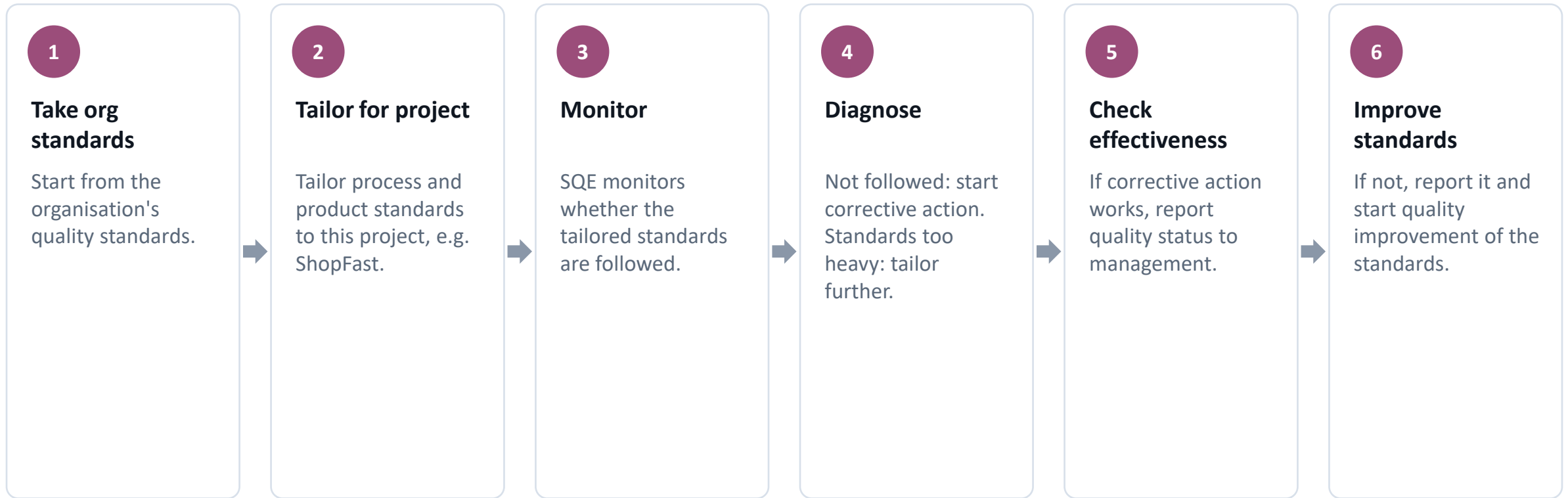
Apply everywhere

Quality improvement starts from what already works well, and spreads those practices. Non-conformances trigger corrective action or a change to the standard itself.

Quality is free, if it is everyone's responsibility.

Quality improvement: the creative approach

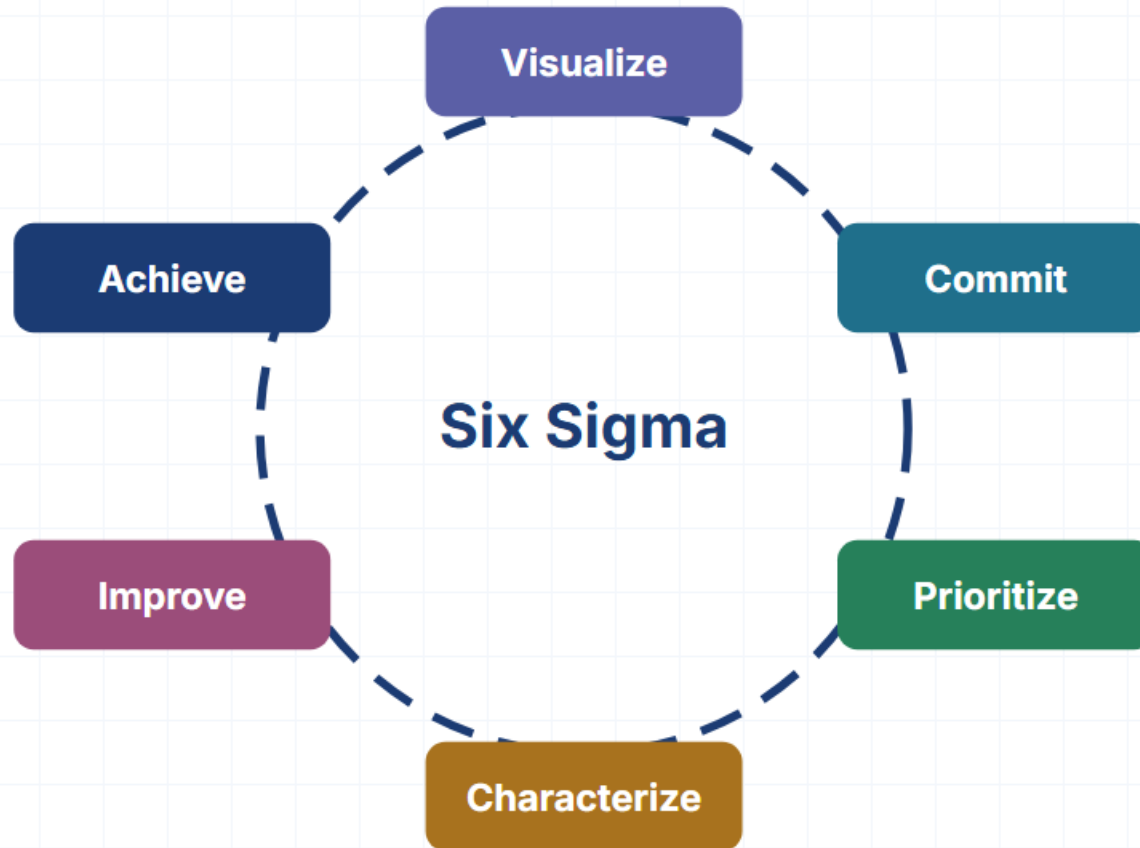
Find good products, study how they were made, and generalise those processes so they apply everywhere



The loop feeds project lessons back into the organisation's standards, so each project makes the next one better.

Quality improvement: the Six Sigma wheel

An improvement cycle driven by data and stakeholder commitment



Visualize: understand how it works now and imagine the future

Commit: get stakeholder commitment to change

Prioritize: choose incremental improvements

Characterize: define the current process and the timeline

Improve: design and implement the improvements

Achieve: realise and sustain the results

DMAIC, the common Six Sigma project cycle: Define, Measure, Analyse, Improve, Control.

Building a defect prevention culture

Quality is free, if it is everyone's responsibility

Blameless root cause

Ask what in the process let a defect through, not who typed it. Blame hides the information you need.

Causal analysis

Group defects by cause and fix the top causes first. CMMI calls this Causal Analysis and Resolution.

Share lessons

Publish short post-mortems across projects so one team's mistake is not repeated by another team.

Standards that fit

Tailor standards so people can follow them. Excessive standards invite quiet non-compliance.

Measure and feed back

Show teams their own defect trends and recognise real improvement, such as fewer escaped defects.

Early involvement

Bring testers and SQE into requirements reviews, where most fielded errors begin.

Early warning: death march projects

Symptoms listed in Leveson's notes, and Yourdon's four types of problem project

Symptoms to watch

- Feature (scope) creep
- Thrashing: much activity, little real progress
- Integration problems, and people overwriting each other's source code
- Constant re-estimation of schedule and effort
- Redesign and rewriting during test
- No documentation of design decisions

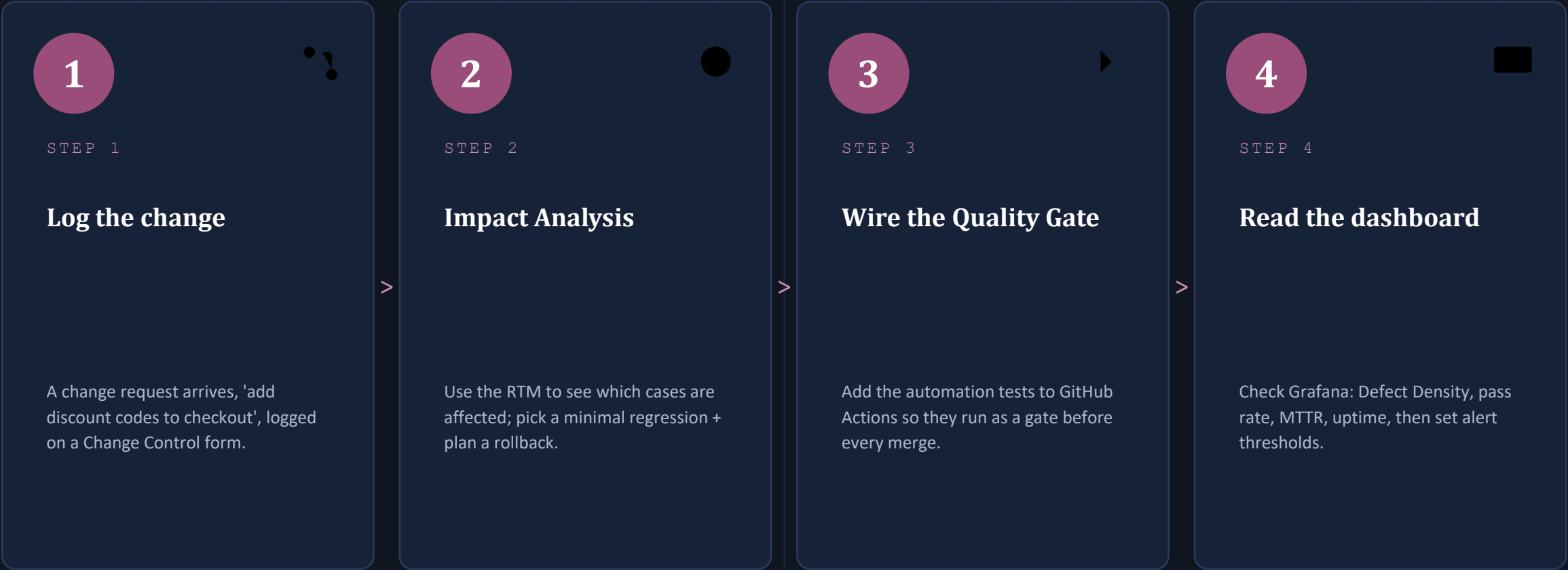
Yourdon's project types

- Mission Impossible: likely to succeed, happy workers
- Ugly: likely to succeed, unhappy workers
- Kamikaze: unlikely to succeed, happy workers
- Suicide: unlikely to succeed, unhappy workers

SQA metrics such as requirements volatility, STR aging and rework during test can reveal these symptoms before the schedule collapses.

Guided lab flow, four steps

Run on the ShopFast checkout artifacts carried from every earlier module.

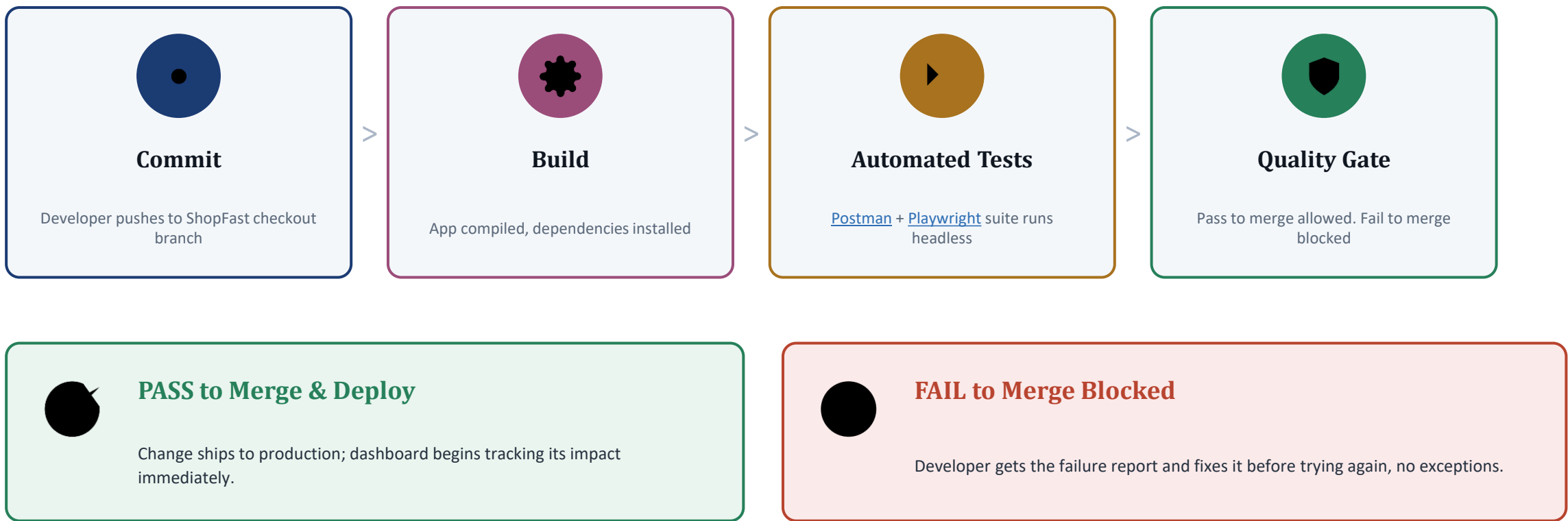


TOOLS FOR THIS LAB

- Change request form
- GitHub Actions
- Grafana / Prometheus / ELK
- RTM

The Quality Gate, no merge without a green run

The Module 5 automation suite now runs automatically on every push, inside [GitHub Actions](#).



Quality Gate dalam CI/CD: tiada merge tanpa larian hijau

CI/CD Quality Gate: no merge without a green run

KRISAv2 Bab 7 7.9: konfigurasi CI/CD dan skrip automasi disimpan dalam repositori. Bab 8: automasi sebelum Laporan Release.



HIJAU

Semua ujian lulus, tiada severity 1 dan 2 terbuka, liputan RTM cukup. Merge dibenarkan dan artifak laporan disimpan.

MERAH

Mana-mana langkah gagal. Merge disekat, pepijat dilog (PPrISA14), pembangun baiki dan tolak semula.

```
lab/shopfast .github/workflows/shopfast-quality-gate.yml job: quality-gate
```

Real run: GitHub Actions quality gate

github.com/FTH-ABR/sqa-krisa-bengkel/actions, job quality-gate

Platform Solutions Resources Open Source Enterprise Pricing

Search

Sign in

Sign up

FTH-ABR / sqa-krisa-bengkel

Public

Notifications

Fork 0

Star 0

<> Code

Issues

Pull requests

Actions

Projects

Security and quality

Insights

ShopFast quality gate

Kit makmal ShopFast: aplikasi, ujian API/UI, quality gate CI; rajah k... #1

Sign in to view logs

Summary

All jobs

quality-gate

Run details

Usage

Workflow file

Triggered via push 5 minutes ago

Status

Total duration

Artifacts

FTH-ABR pushed 9a38342 main

Success

43s

1

shopfast-quality-gate.yml

on: push

quality-gate

40s

Annotations

1 warning

quality-gate

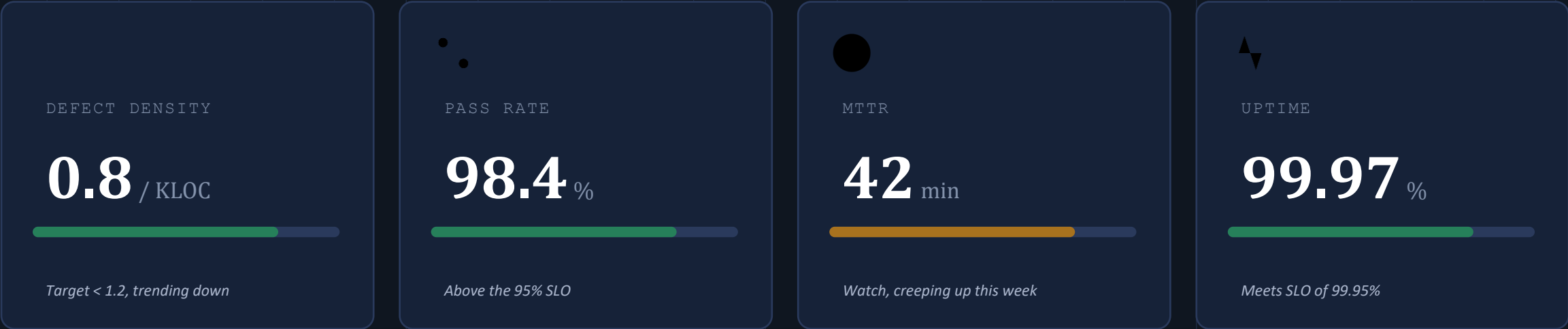
Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, actions/setu...

Show more

Artifacts

Reading the Grafana dashboard

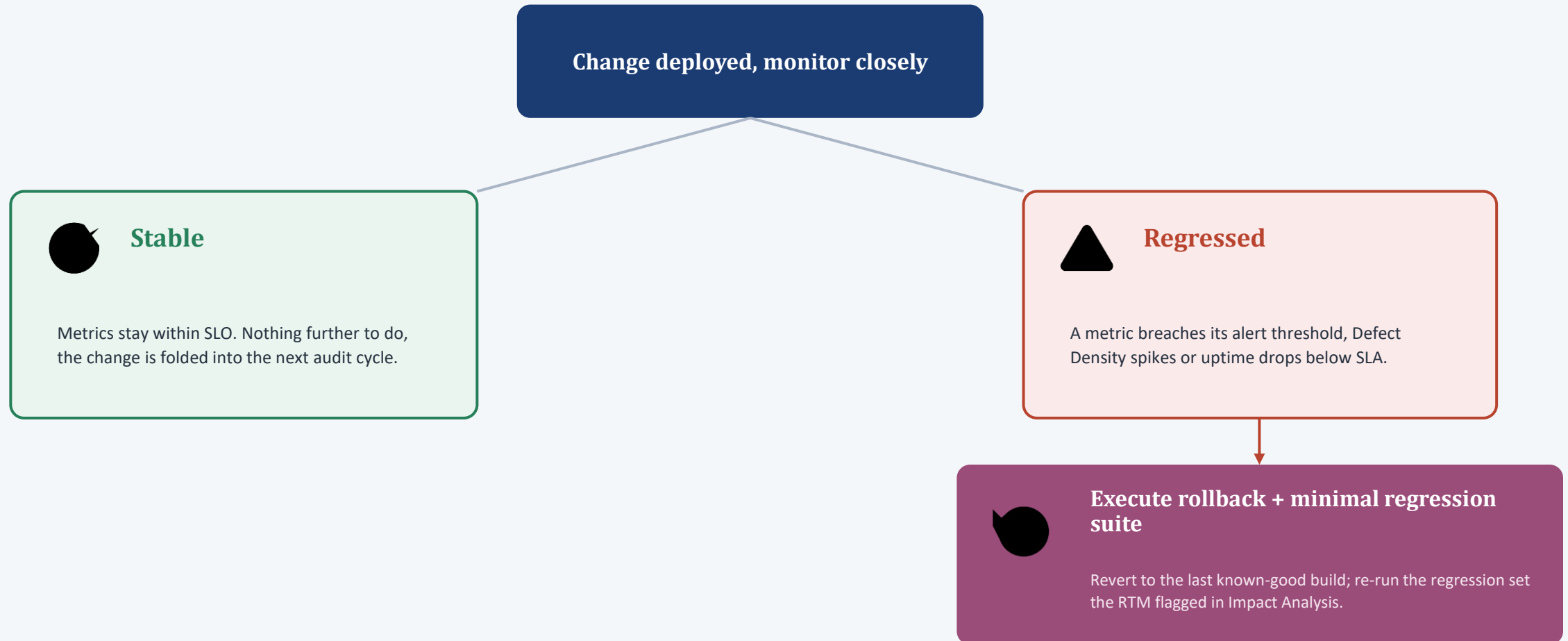
Four numbers tell you whether ShopFast checkout is healthy right now.



Setting alert thresholds: MTTR above 60 min pages the on-call engineer. Uptime under 99.9% (the SLA floor) triggers an incident review. Defect Density above 1.2/KLOC blocks the next release train automatically.

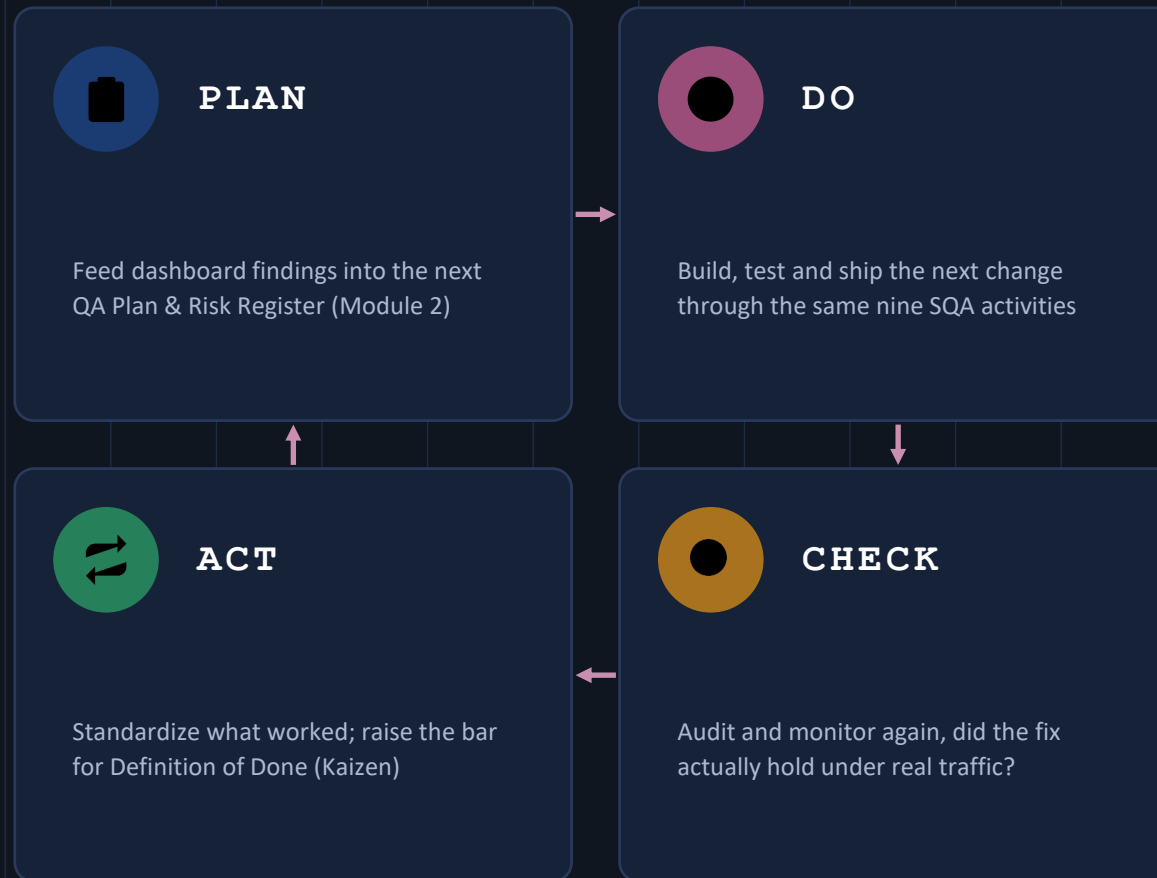
If it breaks, rollback and regression planning

Planned before the change ships, not improvised after it fails.



PDCA & Kaizen, closing the loop back to Module 1

Monitoring doesn't end at the dashboard. What it finds becomes next cycle's quality plan.

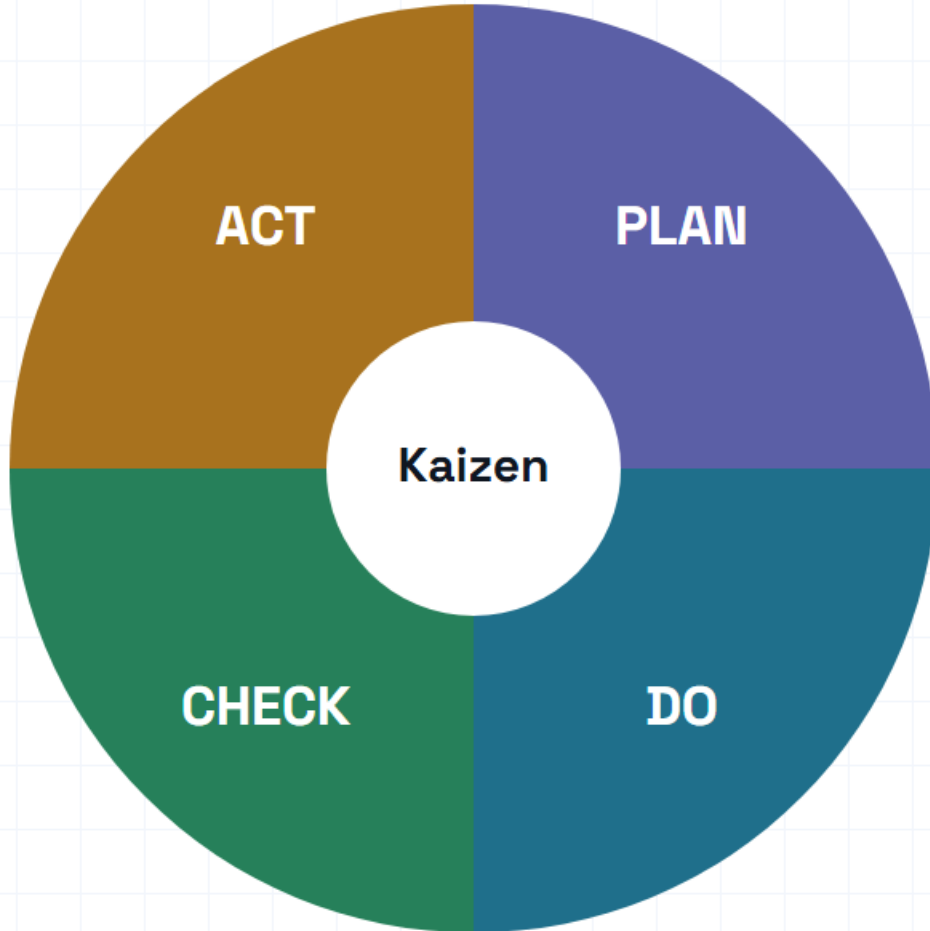


Kaizen mindset: every capstone and every real project restarts this circle, the goal is never a perfect release, only a slightly better next one.

PDCA dan Kaizen: kualitas yang tidak pernah selesai

PDCA and Kaizen: quality is never finished

Setiap retrospektif menghasilkan satu tindakan penambahbaikan yang boleh diukur



Plan: sasaran metrik (contoh: defect leakage bawah 5%), pelan ujian dan risiko.

Do: laksana perubahan kecil, contoh tambah ujian BVA pada kuantiti.

Check: baca papan pemuka, bandingkan SLI dengan SLO, RCA pepijat bocor.

Act: standardkan yang berjaya ke templat dan Quality Gate, ulang kitaran.

Tools & KRISA 2.0 alignment

TOOLS FOR THIS MODULE



Change request form

Structured intake for every proposed change



[GitHub Actions](#)

CI pipeline running the Quality Gate



[Grafana](#) / [Prometheus](#) / ELK

Dashboards, metrics storage & log search



RTM



Reference note: KRISA 2.0 is currently at Beta 2026 status. Deliverable and template numbering is still being refined by JDN, confirm the current version with your agency before formal submission.

KRISA 2.0 (JDN) MAPPING

[Ch.8](#) · DevOps Alignment

Automation, CI/CD, monitoring; KRISA and [PPriSA](#) and DevOps mapping; Change Log

Deliverable: Continuous quality pipeline & metrics

Cross-cutting: Change & Risk Control Management

One of KRISA's key success factors, this module is where it's operationalized.

Deliverable: Change Log; Impact Analysis record

Together with Module 7's [Ch.6-7](#) evidence, this closes KRISA's full Planning-to-Sustain lifecycle for the ShopFast release.

Abbreviations used in this module (1 of 6)

Every short form on these slides, explained in plain words. Keep this page handy during the lab.

Abbreviation	Stands for	What it means in plain words
AI	Artificial Intelligence	Software that can answer questions and write text or code by learning patterns from large amounts of data.
API	Application Programming Interface	A doorway that lets one program ask another program for data or an action, without using a screen.
APM	Application Performance Monitoring	Tools that watch a live system and show how fast and healthy it is.
BVA	Boundary Value Analysis	Testing at the edges of the allowed values, such as 0, 1, 10 and 11, because mistakes often hide there.
CAB	Change Advisory Board	A group that reviews planned changes to a live system and advises whether they should go ahead.
CAPA	Corrective and Preventive Action	Fixing a problem that has happened, then changing the process so it does not happen again.
CCB	Change Control Board	The group that approves, rejects or defers requests to change an approved baseline.
CI	Continuous Integration, or Configuration Item	Two meanings. In DevOps, every code change is built and tested automatically. In configuration management, it is any controlled item such as a document or code file.

Abbreviations used in this module (2 of 6)

Every short form on these slides, explained in plain words. Keep this page handy during the lab.

Abbreviation	Stands for	What it means in plain words
CI/CD	Continuous Integration and Continuous Delivery	An automated pipeline that builds, tests and prepares every change for release.
CMMI	Capability Maturity Model Integration	The newer model that replaced CMM. Used to improve and appraise how an organisation works.
CR	Change Request	A written request to change something already agreed, stating the reason and the impact.
DMAIC	Define, Measure, Analyse, Improve, Control	The five-step project cycle used in Six Sigma improvement work.
ECP	Engineering Change Proposal	A formal proposal to change an agreed baseline, usually with its cost and schedule impact.
ELK	Elasticsearch, Logstash and Kibana	A popular set of tools to collect, search and chart system logs.
EP	Equivalence Partitioning	Grouping inputs that behave the same way, then testing one value from each group.
ID	Identifier	A unique label, such as REQ-CHK-01, so that each item can be traced.

Abbreviations used in this module (3 of 6)

Every short form on these slides, explained in plain words. Keep this page handy during the lab.

Abbreviation	Stands for	What it means in plain words
IEEE	Institute of Electrical and Electronics Engineers	A professional body that publishes software engineering standards such as IEEE 730 and IEEE 829.
ISO	International Organization for Standardization	Publishes international standards such as ISO 9001 and ISO/IEC 25010.
JDN	Jabatan Digital Negara	The Malaysian government department that publishes KRISA and PPrISA.
KLOC	Thousand Lines of Code	A size measure used in rates such as defects per KLOC.
KRISA	Kejuruteraan Sistem Aplikasi Sektor Awam	The Malaysian public sector guide for engineering application systems. KRISAv2 is the 2026 beta edition.
MIT	Massachusetts Institute of Technology	A university in the United States. Source of the Leveson course notes.
MTTR	Mean Time To Repair	The average time needed to get a failed system working again. Lower is better.
PAT	Provisional Acceptance Test	The KRISA acceptance test done before provisional acceptance of the system, including performance checks.

Abbreviations used in this module (4 of 6)

Every short form on these slides, explained in plain words. Keep this page handy during the lab.

Abbreviation	Stands for	What it means in plain words
PDCA	Plan, Do, Check, Act	A four-step loop for continuous improvement.
PDF	Portable Document Format	A document file that looks the same on every computer.
PPrISA	Panduan Pengurusan Projek ICT Sektor Awam	The Malaysian public sector guide for managing ICT projects. Codes such as PPrISA05 are its templates.
PR	Pull Request	A request on GitHub to review and merge a code change into the main branch.
PS	PowerShell prompt	When a slide shows PS>, type that command in Windows PowerShell.
QA	Quality Assurance	Preventing defects by setting up good processes and checking that people follow them.
REQ	Requirement	Prefix in requirement IDs such as REQ-CHK-01. CHK stands for checkout.
RM	Ringgit Malaysia	Malaysian currency.

Abbreviations used in this module (5 of 6)

Every short form on these slides, explained in plain words. Keep this page handy during the lab.

Abbreviation	Stands for	What it means in plain words
RTM	Requirements Traceability Matrix	A table linking each requirement to its design, tests and defects, so nothing is missed.
SCCB	Software Configuration Control Board	The board that controls changes to software baselines, including which bug fixes go in.
SCM	Software Configuration Management	Tracking every version and change so the right items are built and released.
SIT	System Integration Test	Testing that the parts of a system, and the systems it connects to, work correctly together.
SLA	Service Level Agreement	The promise made to users, such as 99.5 percent uptime each month.
SLI	Service Level Indicator	The actual measurement, such as the percentage of checkouts that succeed.
SLO	Service Level Objective	The internal target for an indicator, set tighter than the SLA.

Abbreviations used in this module (6 of 6)

Every short form on these slides, explained in plain words. Keep this page handy during the lab.

Abbreviation	Stands for	What it means in plain words
SQA	Software Quality Assurance	All the planned work that gives confidence the software meets its quality requirements.
SQE	Software Quality Engineer	The person who runs quality checks and audits and witnesses formal tests.
SRS	Spesifikasi Keperluan Sistem	The System (or Software) Requirements Specification: what the system must do. KRISA deliverable D03.
STR	Software Trouble Report	The form that records a bug from the moment it is found until it is closed.
UAT	User Acceptance Test	Users try the system to confirm it meets their needs before they accept it.
UI	User Interface	The screens, buttons and forms that people use.
ZAP	Zed Attack Proxy	A free OWASP tool that scans web applications for security weaknesses.

References and templates for this module

Every line opens the official JDN document at the exact page. Templates download as Word or PDF.

WHERE IT IS IN THE GUIDES

- Bab 8, Jadual 8.1 Pemetaan Aktiviti KRISA ke DevOps
page 6 · KRISAv2 Bab 8 Penyelarasan DevOps
- Bab 8, 8.3.3 Penyelarasan Pengurusan Perubahan dan Risiko
page 9 · KRISAv2 Bab 8 Penyelarasan DevOps
- Bab 7, 7.9 Repositori dan Serahan Kod Sumber (CI/CD)
page 17 · KRISAv2 Bab 7 Fasa Pelaksanaan
- Bab 1, 1.6.5 Pengurusan Kawalan Pindaan
page 22 · KRISAv2 Bab 1 Perancangan
- PPrISA 2.0, 4.2.3 c) Pengurusan Pindaan (CR)
page 73 · PPrISA 2.0 Versi Beta Feb 2025
- PPrISA 2.0, 5.4.2 Pemantauan dan laporan status
page 91 · PPrISA 2.0 Versi Beta Feb 2025

TEMPLATES TO USE

- D18
Laporan Serahan Sistem

WORD

PDF
- PPrISA10
Borang Pindaan

WORD

PDF
- PPrISA11
Log Penyelesaian Pindaan

WORD

PDF
- PPrISA16
Laporan Status Projek (Kumpulan)

WORD

PDF
- PPrISA17
Laporan Status Projek (Keseluruhan)

WORD

PDF

Sustaining quality means never really finishing



Every change, however small, earns an Impact Analysis before it earns a merge.



Logs, metrics and traces, together, are what 'observability' actually means.



A rollback plan written before deployment is worth ten written after an incident.



PDCA turns this module's dashboard findings into next cycle's Module 1 quality mindset.

NEXT to

Capstone Exercise & Course Wrap-Up

16:30-17:30 · One team, one enterprise case study, all nine SQA activities, requirements to monitoring, then presented live.

M08 · W08 · 15 MINUTES · Hari 2, 16:10 hingga 16:25

Hands-On SQA-AI Assistant Workshop

AI change request, impact analysis and CI review

Raise a PPrISA10 change request for a new promo code, let the AI trace the impact through the RTM and test cases, and review the CI quality gate workflow.

TOOL

opencode 1.18 in the lab folder · model: Big Pickle via OpenCode Zen (free, no API key) · synthetic ShopFast data only

[Step-by-step guide \(web\)](#)[Workshop slides \(W08\)](#)[Setup guide W00](#)[opencode docs](#)

AI change request, impact analysis and CI review

STEPS

- 1 Run `/change-request` for promo code RAYA26
- 2 Run `/impact-analysis` (Plan agent)
- 3 Ask the Plan agent to review the GitHub Actions workflow
- 4 Agree the minimal regression set and rollback trigger

OUTPUT TO SAVE

`outputs/m08-change-request.md`, `outputs/m08-impact-analysis.md`

TYPE THIS (opencode> in opencode, PS> in PowerShell)

```
opencode> /change-request Tambah kod promo RAYA26 (diskaun 26 peratus, sah 1 hingga 31 Mac 2027) pada checkout ShopFast
```

```
opencode> /impact-analysis RAYA26
```

```
opencode> Semak ../.github/workflows/shopfast-quality-gate.yml sebagai quality gate. Nyatakan apa yang disemak, apa yang tiada (contoh imbasan keselamatan atau ambang liputan) dan 3 cadangan. Jangan tulis fail.
```

HUMAN CHECK BEFORE YOU ACCEPT IT

- RAYA26 at 26 percent breaks the 25 percent cap in REQ-CHK-03: did the AI notice?
- The regression set is minimal and traced to REQ-CHK IDs
- Rollback has a clear trigger and an owner

[Open step-by-step guide](#)

[Module hub](#)

Pembantu AI dalam SQA: AI mendraf, manusia bertanggungjawab

AI assistant in SQA: AI drafts, humans remain accountable

Aliran kerja setiap bengkel Hands-On SQA-AI Assistant Workshop menggunakan opencode

1

Konteks

@SRS, @templat,
AGENTS.md

2

Prompt

arahan /review-srs

3

Draf AI

opencode + model

4

Semak manusia

banding dengan templat
KRISA/PPrISA

5

Simpan bukti

outputs/ + git commit

Guna AI untuk

Draf pelan dan laporan, cadang kes ujian
EP/BVA, jana skrip ujian, ringkaskan log, cari
jurang RTM.

Jangan

Tampal data sebenar agensi, terima output
tanpa semak, biar AI reka ID keperluan, kongsi
sesi awam.

Bukti audit

Simpan prompt, output dan semakan dalam git
supaya jejak audit lengkap (Modul 7).